Modeling Diversity Decay in Synthetic Retraining

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Abstract

As synthetic data generated by AI models increasingly populates the web, it is inevitably recycled into future training sets, creating a recursive synthetic retraining process. This is known to cause model collapse, where data quality progressively deteriorates and the generated distribution becomes narrower. Although there have been extensive studies on this phenomenon, ways to mitigate it, as well as treating diversity as entangled with fidelity, none have explored whether diversity decays in a precise, predictable trajectory of its own. We investigate this gap through a martingale-based theoretical framework that predicts diversity decay under synthetic retraining and qualitatively support this decay through a high-dimensional Conditional Variational Autoencoder (CVAE) recursively trained on 500 real MNIST images, using a discriminator as an empirical intervention we study. In this setting, we show that diversity — measured most accurately via the Vendi Score — declines steadily under recursive filtering even as the Fréchet Inception Distance (FID) remains stable, revealing a fidelity-diversity dissociation. Hence, under standard diversity metrics such as Coverage, the decline of diversity might not be reliably depicted, but a standalone metric is more suitable. Overall, these results suggest that fidelity-based monitoring alone is insufficient for diversity collapse for recursive synthetic retraining, and that diversity-specific metrics are needed to ensure long-term data quality in generative pipelines.

Keywords

model collapse; synthetic data; diversity decay; recursive retraining; generative models; Vendi Score

Introduction

Synthetic data, data artificially generated by a model or an algorithm, has been used in various fields due to the reduction of costs and resources it can bring (Wood et al. 2021; Chen et al. 2024; Bay et al. 2025). With the rise of synthetic data on the web, generative models such as Large Language Models (LLMs) are increasingly likely to be trained, at least in part, on data generated by earlier versions of themselves. Naturally, many explored the effects of synthetic retraining, where models iteratively train on self-generated data, revealing a phenomenon known as **model collapse**, which is defined as “a degenerative process affecting generations of learned generative models” (Shumailov et al. 2024), whereby each generation is trained on the data polluted by its predecessor’s synthetic output. This causes the model to progressively lose information about the true underlying distribution as the iterations continue, as the low-probability “tail” events disappear due to the finite number of samples.

In response to this phenomenon, a range of strategies have been proposed to slow or avoid model collapse. **Data accumulation**, in which real data is accumulated over generations rather than replaced across generations, has been shown both theoretically and empirically to bound test error where simple replacement causes it to grow unboundedly (Gerstgrasser et al. 2024). **Real-data mixing** at a fixed proportion each generation has also been shown to avoid collapse under appropriate conditions: Bertrand et al. (2024) prove that collapse can be avoided in a replacing-data setting when the initial generative model approximates the true distribution well and the real-data proportion is sufficiently large. Another one is **the use of verifiers**, where the low-quality synthetic samples are filtered using an external verifier

(also called the discriminator) before they are used for retraining. Yi et al. (2026) introduce the "Bias–Variance Tradeoff," where verifier-guided retraining yields near-term improvement but ultimately converges to the verifier's own biased "knowledge center" rather than the true parameter. Our work adopts a similar filtered-retraining setup, but where Yi et al. (2026) study convergence of a point estimate under filtering, we use filtering as a methodological tool to isolate and characterize the trajectory of output *diversity* specifically, independent of the estimator's convergence behavior.

Among the effects of model collapse, the loss of **diversity**, or the range of outputs a model is capable of producing, is of particular concern. Previous works on model collapse and its mitigation strategies have often used the term "quality" to mainly refer to the fidelity or the realism of the data, without separately measuring diversity as its own axis. While investigating these mitigation strategies under various scenarios, their effectiveness on diversity specifically remains unclear. This is important to note because data diversity can directly affect downstream model performance (Zhu et al. 2025). In addition, the loss of diversity can limit the practical value of a generative model even when the data quality, by conventional measures, remains high, since it does not effectively depict the original distribution. The data generated can be individually polished, but are homogeneous (Doshi and Hauser 2024).

This loss of diversity is also critical in a more real-world context, especially in creating societal challenges. When AI is recursively trained on self-generated data, the output is homogenized, and can produce the so-called "homogenization death spiral," where diversity continuously shrinks across training cycles. This phenomenon can reflect certain biases in AI, consequently leading to societal bias, harming users with less popular preferences (Castro et al. 2025). This narrowing does not affect all groups equally: LLMs have been shown to disproportionately reflect a narrow demographic – predominantly Western, high-income, and highly educated populations – while marginalizing non-Western cultures and underrepresented groups within Western societies (Daryani et al. 2025). By mid-2025, 35% of newly published websites were found to have been generated or assisted by AI, showing how significant this issue is (Dolezal et al. 2026).

Hence, it is crucial to examine how diversity decays in synthetic retraining. While previous works have established that model collapse occurs and measured its aggregate effects on model performance (Shumailov et al. 2024; Dohmatob et al. 2024), none have characterized the specific functional form this decay takes as generations progress, nor examined whether it follows a consistent, derivable pattern across settings. Knowing the shape of the decay, rather than just the fact that it will occur, would allow for a practical means of anticipating collapse before it becomes empirically visible, potentially allowing practitioners to intervene, such as adjusting the proportion of real data mixed into training, in a more efficient manner before diversity loss becomes too severe. Thus, a natural question arises:

Does diversity decay under recursive synthetic retraining follow a predictable functional form?

Our contributions are as follows:

- We derive that variance in a recursively retrained Gaussian generative process decays almost surely to zero even though its expected value remains constant across generations, and show through simulation that combining a verifier with real-data mixing yields a stable, nonzero equilibrium.
- We qualitatively support this decay behavior in a realistic, higher-dimensional setting, showing that within-class output diversity in a verifier-filtered CVAE trained recursively on MNIST declines substantially (by over 60% in Vendi Score) across 50 generations.
- We show that this diversity collapse is not reliably captured by standard quality-oriented metrics: the Coverage improves over the same generations in which diversity and held-out reconstruction loss both substantially worsen, while FID, an aggregate metric, shows no consistent trend in

either direction, demonstrating that diversity loss can proceed undetected by commonly used evaluation metrics.

Methods

Recursive Retraining Process

We consider a generic recursive retraining process common to the model collapse literature. Let p_0 denote the true data distribution, and let $\hat{p}_0$ denote an initial generative model trained on n_0 real samples drawn from p_0. At each generation $t = 1, \ldots, T$, the current model $\hat{p}_{t-1}$ produces a batch of N synthetic samples. If a verifier is used, each sample is scored by an external verifier V, and only the top fraction $q \in (0,1]$ of samples, as ranked by V, is retained, forming a filtered training set S_t; if no verifier is used, S_t consists of all N generated samples. A new model $\hat{p}_t$ is then trained from scratch on S_t alone, without access to $\hat{p}_{t-1}$'s training data or any earlier generation's samples. This matches the *replace* setting studied in prior work (Shumailov et al. 2024; Gerstgrasser et al. 2024), in which each generation's model is trained exclusively on its immediate predecessor's output.

The subsections below are instances of this general process in two settings: a simple, one-dimensional Gaussian model that serves as a simulation for theoretical analysis, and a Conditional Variational Autoencoder (CVAE) trained recursively on MNIST, which serves as a higher-dimensional empirical test of the same underlying dynamics.

Measuring Diversity

We define diversity as the range of distinct outputs a model is capable of producing for a given class or category, not the fidelity or realism of any individual output. We therefore restrict our definition of diversity to *within-class variance*: for the Gaussian setting, this is the variance σ_t^2 of the estimated distribution at generation t; for the MNIST setting, this is the variation in stroke width, slant, and handwriting style among generated samples of the same digit class. In the Gaussian setting, where the data-generating process is fully known, diversity is measured directly via the estimated variance $\hat{\sigma}_t^2$.

For the CVAE experiment, we measure diversity mainly through three established metrics for measuring image generation and distribution quality.

Fréchet Inception Distance (FID): A widely-used metric introduced by Heusel et al. (2018), the FID measures how well the generated images resemble the real data, capturing both fidelity and diversity. The FID compares the mean and covariance of embedded features between the reference dataset and an evaluation dataset:

$$\mathrm{FID} = \|\mu_{real} - \mu_{gen}\|_2^2 + \mathrm{Tr}\left(\Sigma_{real} + \Sigma_{gen} - 2\left(\Sigma_{real}\Sigma_{gen}\right)^{\frac{1}{2}}\right)$$

where μ_{real}, Σ_{real} and μ_{gen}, Σ_{gen} are the mean and covariance of the real and generated sample embeddings respectively, and $\mathrm{Tr}(\ldots)$ is the trace of the matrix. Unlike the standard formulation, which embeds images using an ImageNet-pretrained InceptionV3 Network, we embed images using a 64-dimensional encoder of a CVAE trained from scratch on real MNIST digits. This is because applying a large-scale Image-Net trained network to a simple MNIST dataset of 28x28 pixels is unsuitable. In addition, the encoder better preserves within-class variation (e.g., stroke, width, slant) relevant to evaluating diversity in this setting. The higher the FID, the lower the image quality is.

Coverage: Improving on earlier Precision/Recall metrics, Naeem et al. (2020) introduced Density and Coverage by being more robust to outliers while separately capturing fidelity (density) and diversity (coverage), in contrast to aggregate metrics such as FID that conflate the two into a single score. Since our experiment emphasizes the diversity aspect, we measure the Coverage, which quantifies the fraction of real samples whose neighborhood contains at least one generated sample:

$$\text{coverage:} = \frac{1}{N}\sum_{i=1}^{N} 1_{\exists\ j \text{ s.t. } Y_j \in B(X_i, \text{NND}_k(X_i))}$$

where $B(x, r)$ is the ball of radius r centered at x; $\text{NND}_k(X_i)$ is the distance from real sample X_i to its k-th nearest neighbor among the real samples; X_i and Y_j denote the i-th real sample and j-th generated sample respectively; and N is the number of real samples. In our setting, $k = 10$. For Coverage, we use the same embedder used for measuring FID.

Vendi Score: The Vendi Score utilizes the ecological definition of diversity to measure diversity in AI-generated images, and does not require a reference dataset unlike the previously mentioned metrics. It is defined as:

$$\text{VS} = \exp\left(-\sum_{i=1}^{n} \lambda_i \log \lambda_i\right)$$

where n is the number of generated samples, $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $\frac{K}{n}$, and K is the $n \times n$ similarity matrix with entries $K_{ij} = k(Y_i, Y_j)$ for a chosen similarity kernel $k(\cdot,\cdot)$ satisfying $k(x, x) = 1$; we use as k the cosine similarity between the outputs of the same embedding function used for the two previously mentioned metrics above. We compute the Vendi Score within each digit class separately, then macro- average them across the 10 classes.

The Role of Filtering

Recursive retraining on unfiltered synthetic data conflates two distinct effects: degradation in sample fidelity, in which generated outputs become less realistic or accurate, and degradation in diversity, in which generated outputs become less varied. Both of these effects can occur simultaneously and can affect the diversity measurements severely. We aim to mitigate this through the use of a verifier, which selects high-fidelity samples, allowing us to observe the trend of diversity specifically without the fidelity degrading alongside it. This setup follows the verifier-based synthetic retraining framework of Yi et al. (2026); however, our work simply employs the verifier as a tool to isolate diversity.

Theoretical Setup

As a simpler case of the general process described above, we look at a one-dimensional Gaussian generative process, where each generation estimates a mean and variance from a finite sample drawn from its predecessor's distribution.

Let the true data distribution be $\mathcal{N}(\mu_0, \sigma_0^2)$. At generation $t = 0$, an initial model estimates $\hat{\mu}_t, \hat{\sigma}_t^2$ from n i.i.d. samples drawn from the true distribution. At each subsequent generation $t \geq 1$, the model draws n i.i.d. samples from $\mathcal{N}(\hat{\mu}_{t-1}, \hat{\sigma}_{t-1}^2)$, which is its predecessor's estimated distribution, and computes new estimates using the standard unbiased sample mean and variance. The real data is not used after generation 0, as only the current generation's distribution matters for the next generation, $\mathcal{N}(\hat{\mu}_{t+1}, \hat{\sigma}_{t+1}^2)$.

By the standard theory of Gaussian sampling, the sample variance satisfies

$$\hat{\sigma}_t^2 = \hat{\sigma}_{t-1}^2 \cdot R_{t-1}, R_{t-1} \sim \frac{\chi_{n-1}^2}{(n-1)},$$

with R_{t-1} independent of $\hat{\sigma}_{t-1}^2$ and $\mathbb{E}[R_{t-1}] = 1$. Unrolling this recursion gives

$$\hat{\sigma}_t^2 = \hat{\sigma}_0^2 \prod_{k=0}^{t-1} R_k.$$

Variance Decays Almost Surely

Proposition 1 (Martingale property). $(\hat{\sigma}_t^2)_{t\geq 0}$ *is a martingale:* $\mathbb{E}[\hat{\sigma}_{t+1}^2 \mid \hat{\sigma}_t^2] = \hat{\sigma}_t^2$*, and consequently* $\mathbb{E}[\hat{\sigma}_t^2] = \sigma_0^2$ *for every* t*.*

Proof. Immediate from $\mathbb{E}[\hat{\sigma}_{t+1}^2 \mid \hat{\sigma}_t^2] = \hat{\sigma}_t^2 \cdot \mathbb{E}[R_t] = \hat{\sigma}_t^2$, and induction on the recursion above.

One might believe that Proposition 1 suggests diversity should simply fluctuate around σ_0^2 forever. Our main theoretical result shows this is false.

Theorem 1 (Almost sure collapse). $\hat{\sigma}_t^2 \to 0$ *almost surely as* $t \to \infty$*.*

Proof. Since $(\hat{\sigma}_t^2)_{t\geq 0}$ is a non-negative martingale, Doob's Martingale Convergence Theorem (Doob 1953) guarantees it converges almost surely to a finite limit. Taking logarithms of the recursion above, $log\hat{\sigma}_t^2 = log\hat{\sigma}_0^2 + \sum_{k=0}^{t-1} log R_k$, a sum of i.i.d. terms. By the Strong Law of Large Numbers, $\frac{1}{t}\sum_{k=0}^{t-1} log R_k \to \mathbb{E}[log R_0]$ almost surely. Since log is strictly concave and $\mathbb{E}[R_0] = 1$, Jensen's inequality gives $\mathbb{E}[log R_0] < 0$ strictly, so the sum diverges to $-\infty$ almost surely, and hence $\hat{\sigma}_t^2 \to 0$ almost surely.

The two results are not in contradiction, though. The average of $\hat{\sigma}_t^2$ across all possible realizations remains σ_0^2 forever, while almost every individual case that is actually observed converges to zero.

Gaussian Toy Model

We simulate the recursive retraining process described above with $\mu_0 = 0$, $\sigma_0 = 1$, and $n = 50$ samples drawn per generation, run for 200 generations across 500 independent trials to characterize the distribution of outcomes. At each generation, 50 samples are drawn from the previous generation's estimated distribution, and new mean and variance estimates are computed using the methods outlined above.

To model real-world circumstances, we additionally run experiments with the use of a verifier, mix of a fixed fraction $\rho = 0.1$ of the real data at every generation's n samples, or both. The verifier rejects any generated sample falling outside a constant absolute distance $r = 1$ from the current estimated mean, independent of the current estimated variance $\hat{\sigma}_{t-1}^2$, before the remaining samples are used to compute the next generation's estimates.

CVAE/MNIST Experiment

To evaluate whether the decay behavior observed in the Gaussian toy model is kept in a higher-dimensional, realistic generative setting, we run the recursive retraining process described above using a Conditional Variational Autoencoder (CVAE) trained on MNIST, with an architecture heavily adapted from Yi et al. (2026).

Mirroring the conditions tested in the Gaussian toy model, we run the CVAE/MNIST experiment in three settings: without a verifier, with a verifier, and with both a verifier and real-data mixing. In the unfiltered

setting, no discriminator is trained, and each new CVAE is trained on all 10,000 synthetic images produced by its predecessor.

For the filtered setting, a discriminator is trained at each generation to distinguish real MNIST images from the current generation's synthetic output; this discriminator serves as the verifier described in The Role of Filtering section above. Then, the current generator produces candidate images, which the verifier ranks within each digit class; the top 10% are retained, and generation continues until 10,000 samples have been accepted (1,000 per digit class), ensuring class balance is preserved by construction to accurately measure diversity. Retaining 10,000 samples at a 10% acceptance rate requires drawing roughly 120,000 candidate images per generation. The verifier is not a part of the CVAE's training objective; no gradient from the discriminator ever updates the generator. Finally, A new CVAE is trained on this filtered data alone, with no data retained from earlier generations and no real data used after generation 0.

In the mixing setting, we retain the verifier and additionally mix a fixed fraction $\rho = 0.1$ of real data into every generation's training set: each generation trains on 9,000 verifier-filtered synthetic images together with 1,000 real MNIST images, preserving both the total training set size of 10,000 and class balance by construction (900 synthetic and 100 real images per digit). As in the Gaussian setting, the mixing fraction is defined relative to each generation's training set rather than relative to the number of real images used at generation 0, so each generation receives 1,000 real images, twice the 500 used to train the initial model. The real images are drawn fresh from the full 60,000-image MNIST training set at every generation, so the same images are not repeatedly shown to successive models, and are binarized to match the format of the synthetic samples, which are drawn as Bernoulli samples from the decoder.

For all of the three settings above, an initial CVAE with a 20-dimensional latent space is trained on 500 real MNIST images. Labels are concatenated in two ways: onto the flattened convolutional features before the layer that produces the mean and log-variance, and onto the sampled latent code before the decoder's first layer. The encoder has two convolutional layers (1 to 32 channels, then 32 to 64 channels) using 4×4 kernels and stride 2, downsampling the input from 28×28 to 7×7. The decoder has two transposed-convolutional layers mirroring the encoder to upsample the features back to 28×28. The model is trained to minimize the loss, composed of the reconstruction loss and the KL divergence, using the Adam optimizer with a learning rate of 10^{-3} for 200 epochs, early stopping with patience 5. The training processes are repeated for 50 generations, using a single random seed. At every generation, we compute the diversity and fidelity metrics described above: FID, Coverage, the Vendi Score, and reconstruction loss. In settings involving a discriminator, samples considered "high-quality" are selected, and we measure these metrics on the filtered batch only to minimize the effects of low fidelity, since we are primarily concerned with measuring diversity as its own axis.

Results and Discussion

Gaussian Toy Model

We first verify the theoretical predictions from the Methods section above through direct simulation.

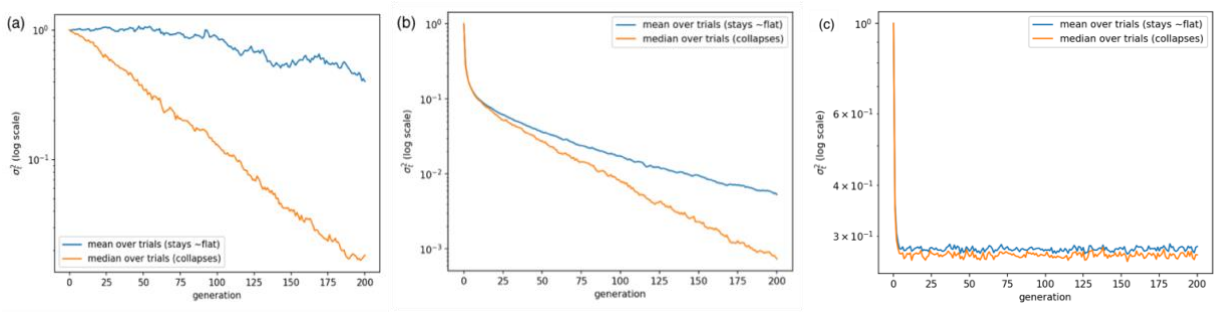


Figure 1. Results of the Gaussian Toy Simulation under three different settings across 500 trials for 200 generations. (a) Mean and median variance without filtering or data-mixing. (b) Mean and median variance with the use of a verifier. (c) Mean and median variance with the use of a verifier and data-mixing.

Figure 1a shows the estimated mean and median variance for 200 generations across 500 trials without filtering or data-mixing. It shows the median variance decaying toward zero, but the mean variance staying relatively flat. This exactly models the theoretical predictions derived in Methods above. The median variance represents the typical, individual runs that actually have variance collapsing, but the mean reflects Proposition 1's guarantee that $\mathbb{E}[\hat{\sigma}_t^2] = \sigma_0^2$ for every generation. A small number of rare, large-variance trials are enough to keep the average flat even as most individual trials are collapsing toward zero. Figure 1b shows the same values but with the use of a verifier. We observe that the median variance drops more smoothly, and the mean now declines alongside it as well. This is because the verifier introduces a systematic reduction in variance at each generation, rather than the unbiased multiplicative factor R_t introduced in Methods above. The discriminator rejects any sample falling outside a fixed absolute distance from the estimated mean, which would break the martingale property of Proposition 1, since $\mathbb{E}[\hat{\sigma}_t^2]$ no longer remains constant at σ_0^2. The verifier's filtering causes the expected variance to decay as well.

Figure 1c shows the mean and median variance across 500 trials under both real-data mixing ($\rho = 0.1$) and the fixed-threshold verifier. Unlike Figure 1b, where mean and median diverged, here both quantities drop sharply over the first several generations before stabilizing together at a shared nonzero equilibrium for the remainder of the run. They both converge to ≈ 0.28. The convergence of mean and median is expected, since filtering breaks the martingale property.

These results directly answer our central research question for the unfiltered case. Diversity decay follows a predictable functional form, specifically geometric decay in expectation over the logarithm of variance, at a rate governed by $\mathbb{E}[logR_0]$. Figure 1a confirms this empirically, as the median variance decays approximately linearly on the log scale across generations, consistent with the exponential form $\hat{\sigma}_t^2 \approx \sigma_0^2 \cdot e^{t \cdot \mathbb{E}[logR_0]}$ predicted by Theorem 1.

CVAE/MNIST Experiment

We next evaluate whether this qualitative decay behavior persists in a higher-dimensional, realistic generative setting.

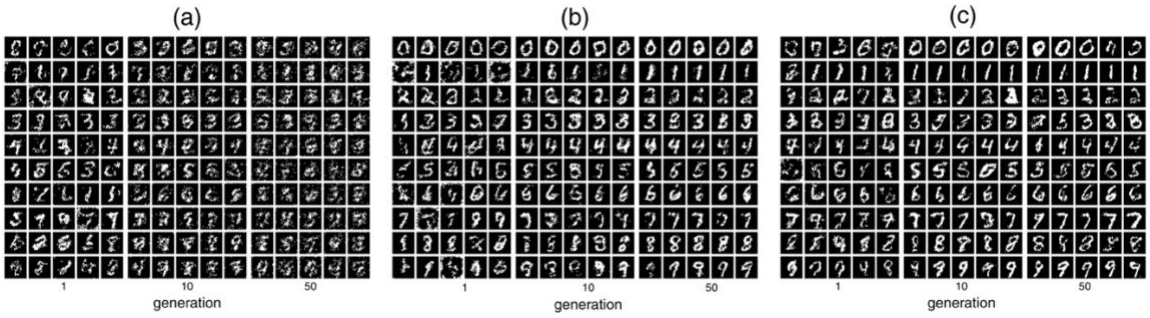


Figure 2. Samples from the CVAE for generations 1, 10, and 50 under each of the three settings. Each panel shows five samples per digit class, one row per class (0–9), generated by sampling $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and decoding conditioned on the digit label (Bernoulli-sampled output). Displayed samples are an arbitrary subset of each batch rather than the highest-scoring ones. (a) Samples drawn from the unfiltered setting. (b) Samples retained after top-10% verifier screening. (c) Samples retained after top-10% verifier screening combined with real-data mixing ($\boldsymbol{\rho} = \mathbf{0.1}$); the real images mixed into training are not shown.

Figure 2 shows representative samples at generations 1, 10, and 50: the unfiltered condition (Figure 2a) visibly degrades across generations, while the filtered (Figure 2b) and filtered-with-mixing (Figure 2c) conditions remain visually coherent.

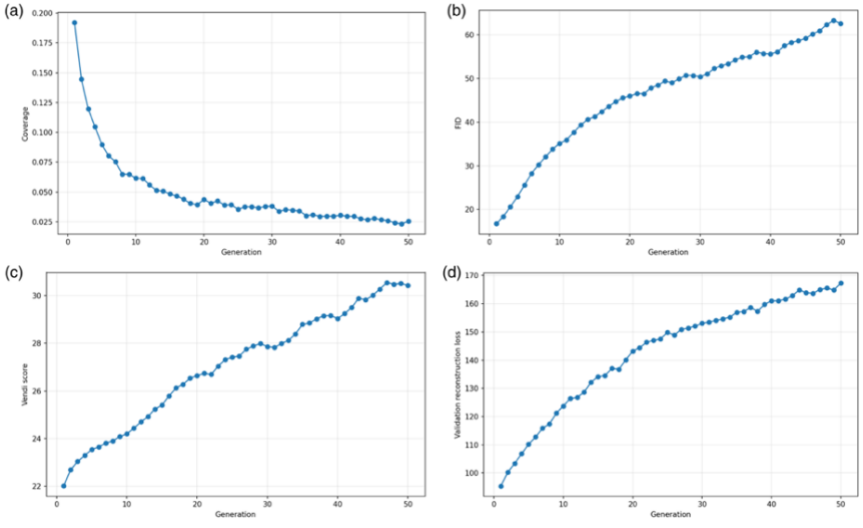


Figure 3. Results of the CVAE experiment under the unfiltered, no-mixing setting for 50 generations. (a) Coverage of the 10,000 synthetic samples. (b) FID score of the 10,000 synthetic samples. (c) Vendi Score of the 10,000 synthetic samples. (d) Validation reconstruction loss on held-out real MNIST digits.

Figure 3 reports the CVAE experiment's behavior under the unfiltered, no-mixing baseline condition. Consistent with the broader model collapse literature, fidelity degrades substantially. FID rises from 16.7 to 62.6 (a 275% increase; Figure 3b), and reconstruction loss on held-out real digits climbs from 95.4 to

167.3 (a 75% increase; Figure 3d). Coverage collapses as well, falling from 0.192 to 0.026 (an 87% decrease; Figure 3a). In contrast, the Vendi Score rises over the same 50 generations, from 22.0 to 30.4 (a 38% increase; Figure 3c). We further explain this result in the Discussion below.

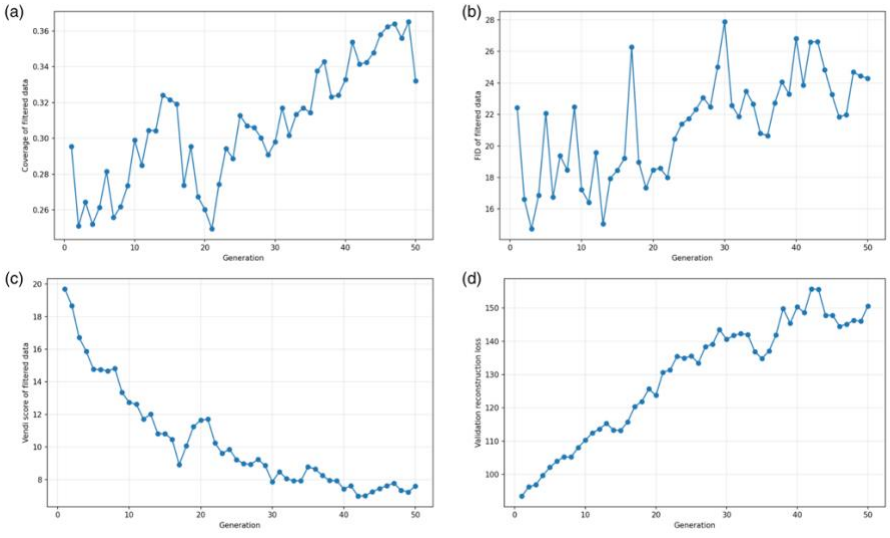


Figure 4. Results of the CVAE experiment under the verifier-filtered setting without real-data mixing for 50 generations. (a) Coverage of the top-10% verifier-filtered synthetic samples. (b) FID score of the top-10% verifier-filtered synthetic samples. (c) Vendi Score of the top-10% verifier-filtered synthetic samples. (d) Validation reconstruction loss on held-out real MNIST digits.

Figure 4 directly demonstrates diversity loss as predicted before. The Vendi Score declined from 19.7 to 7.4 (a 62% decrease) across 50 generations (Figure 4c). Since no real data was involved after generation 0, the diversity kept decaying. Diversity falls sharply over approximately the first 15 generations before visibly slowing and plateauing for the remainder of the run, resembling the exponential decay observed in the median variance curve of Figure 1, though we do not claim a precise quantitative match given the substantially higher dimensionality of this setting and the use of a learned verifier. Figure 4d indicates real degradation is occurring alongside this decline in diversity: the reconstruction loss rises from approximately 93.4 to 150.6, an increase of roughly 61%, over the same 50 generations. This confirms that the model is progressively losing its ability to accurately reconstruct real, unseen MNIST digits, consistent with the model collapse phenomenon discussed in the Introduction above. Yet Figures 4a and 4b show that this diversity loss is not visible in two commonly used evaluation metrics. Coverage rises from approximately 0.30 to 0.33 over the same 50 generations in which the Vendi Score fell by over 62% (Figure 4a). FID fluctuates between approximately 15 and 27 across the run, with no clear upward or downward trend (Figure 4b). Both metrics would suggest that the model is stable or even improving, despite the substantial diversity loss shown in panel (c) and the genuine degradation shown in panel (d).

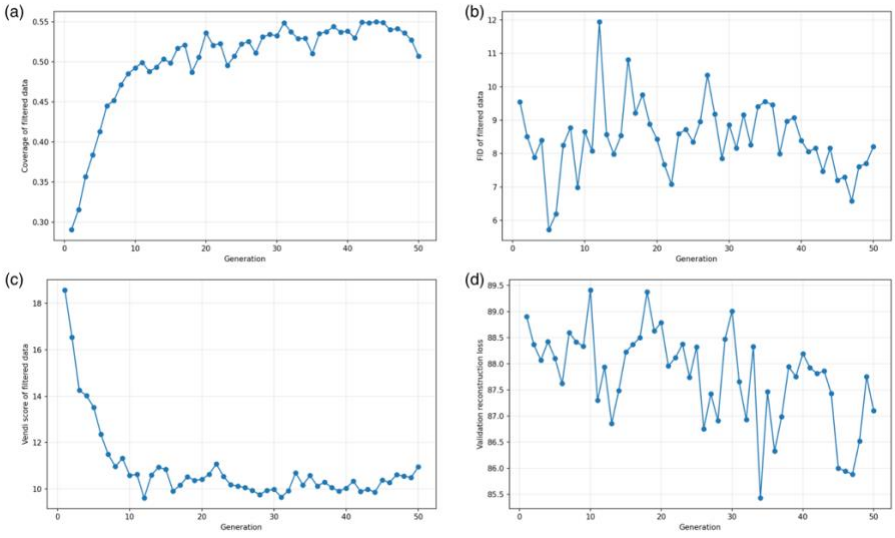


Figure 5. Results of the CVAE experiment under the verifier-filtered setting with real-data mixing ($\boldsymbol{\rho = 0.1}$) for 50 generations. (a) Coverage of the top-10% verifier-filtered synthetic samples. (b) FID score of the top-10% verifier-filtered synthetic samples. (c) Vendi Score of the top-10% verifier-filtered synthetic samples. (d) Validation reconstruction loss on held-out real MNIST digits.

As Figure 5 indicates, the CVAE experiment with both the verifier and real-data mixing yields results similar to the Gaussian simulation. The Vendi Score drops sharply over approximately the first 10 generations, from 18.6 to around 10.5, and then stabilizes for the remainder of the run (Figure 5c). Unlike Figure 4, where the Vendi Score kept declining until generation 50, here it stays at a nonzero level of roughly 10 to 11, for a total decrease of about 41%. This matches Figure 1c, where the variance drops over the first several generations before stabilizing at a nonzero equilibrium. The reconstruction loss also shows the effect of real-data mixing. It stays between roughly 85 and 89 across all 50 generations with no upward trend (Figure 5d), while it rose by 61% in Figure 4. Real-data mixing therefore stops the diversity decline at a nonzero level, as in the Gaussian setting, and also prevents the fidelity degradation seen in Figure 4. As with Figure 4, we do not claim a precise quantitative match, since the Vendi Score is not a direct measure of variance.

The main difference from the Gaussian simulation lies in the other two metrics. Coverage rises from approximately 0.29 to 0.51 (a 75% increase; Figure 5a), a much larger increase than in Figure 4, even though the Vendi Score falls by roughly 41% over the same generations. FID fluctuates between approximately 6 and 12 with no clear trend (Figure 5b). This disagreement is not limited to this setting. In all three CVAE settings, Coverage and the Vendi Score move in opposite directions: in the unfiltered setting, Coverage falls by 87% while the Vendi Score rises by 38%, and in both filtered settings, Coverage rises while the Vendi Score declines. Such a disagreement cannot occur in the Gaussian model, where the data-generating process is fully known and diversity is measured directly through the variance. In the CVAE experiment, diversity can only be estimated through these metrics, and they do not agree with each other. The unfiltered setting is also the only case where the Vendi Score contradicts the Gaussian model in direction. The median variance collapses in Figure 1a, but the Vendi Score rises in Figure 3c, even though the samples in Figure 2a visibly degrade. These results suggest that each metric

responds to more than diversity alone. We discuss the possible reasons behind each metric's behavior in the Discussion below.

Discussion

Prior work on model collapse mitigation has largely evaluated success in terms of fidelity or aggregate test performance (Bertrand et al. 2024; Gerstgrasser et al. 2024; Dohmatob et al. 2024). Our results extend this line of work to diversity specifically. Mixing real data together with the verifier, as done in the simulation, establishes a nonzero equilibrium in variance, but at a value that is much lower than the original variance. The CVAE experiment shows the same pattern: with real-data mixing, the reconstruction loss stays flat across all 50 generations, while the Vendi Score still falls by roughly 41% before plateauing. Collapse in the sense of Theorem 1 is avoided, but a permanent loss of diversity exists. This is easy to overlook when measuring fidelity, but can be visible through a diversity-specific analysis.

The results under filtering also relate to Yi et al. (2026), who show that verifier-guided retraining converges to the verifier's own biased "knowledge center" rather than the true parameter. Our filtered setting points to a similar effect on diversity. Since only the top 10% of samples are kept at each generation, the verifier repeatedly selects what it considers most realistic, and the outputs may narrow toward this preference over time. The retained samples remain visually coherent (Figure 2b), yet the Vendi Score falls by 62%. This matches the observation that generated data can be individually polished, but homogeneous (Doshi and Hauser 2024). The same effect appears in the Gaussian simulation, where the verifier breaks the martingale property and causes the expected variance to decay as well (Figure 1b).

The most practical finding of this work is perhaps the fact that diversity loss can proceed undetected by standard evaluation metrics. In our MNIST experiment with the verifier, Coverage rose over the same generations in which the Vendi Score fell by around 62% and the reconstruction loss rose by roughly 61%, and FID showed no consistent trend. The same disagreement appears in the other two settings, as Coverage and the Vendi Score move in opposite directions in all three. A practitioner monitoring only Coverage or FID during a verifier-filtered recursive retraining process would see no sign of this decline in diversity.

This disagreement can be explained by what each metric measures. The Vendi Score measures how dissimilar the generated samples are from each other and does not require a reference dataset (Friedman and Dieng 2023), so it cannot tell whether that dissimilarity comes from valid variation. In the unfiltered setting, samples gradually lose their digit structure (Figure 2a), and these malformed samples likely spread out in the embedding space, which the Vendi Score counts as higher diversity. Coverage only counts real samples whose neighborhood contains a generated sample (Naeem et al. 2020), so it drops when fidelity collapses in the unfiltered setting. Under filtering, the verifier keeps samples in the dense, typical regions of each digit class, where most real samples lie, so Coverage can rise even as rarer handwriting styles disappear. Since the low-probability tails are the first to be lost in model collapse (Shumailov et al. 2024), Coverage is largely insensitive to this kind of diversity loss. FID, as an aggregate metric (Heusel et al. 2018), can have its diversity losses offset by fidelity gains from filtering, and only rises clearly in the unfiltered setting, where both degrade together. This suggests that a diversity-specific metric such as the Vendi Score should be tracked directly, in addition to a fidelity measure tied to real data such as reconstruction loss, in any pipeline that relies on recursive retraining, since neither type of metric is sufficient alone.

There are several limitations regarding the MNIST experiment worth noting. First, it used one random seed. We do not report variance across multiple seeds, which might lead to different results from different

trials. Second, it was conducted at small scale, with an initial generative model trained on only 500 real MNIST images. Whether the functional form we observe is kept at a larger scale or in other domains, such as text or natural images, remains an open question, so solely relying on this single experiment is insufficient. Third, we tested only one mixing fraction ($\rho = 0.1$) and one filtering fraction (the top 10%), and only in the replace setting. We did not test data accumulation, which Gerstgrasser et al. (2024) show behaves differently from replacement, and a larger real-data proportion may change the equilibrium, since the result of Bertrand et al. (2024) depends on that proportion being sufficiently large. In addition, the mixing setting gives each generation 1,000 fresh real images, twice the number used to train the initial model, so part of the improvement in Coverage and FID under mixing may come from the added real data itself.

There are also limitations in how diversity is measured and modeled. All three image metrics use the same CVAE encoder trained on real MNIST digits rather than InceptionV3, so our FID values are not directly comparable to those in other work. We also measure these metrics on the filtered batch only, so the verifier's selection directly shapes the samples being evaluated. Finally, Theorem 1 covers only the unfiltered Gaussian case. The effects of the verifier and real-data mixing were observed through simulation rather than derived, and the fixed-distance verifier in the Gaussian model works differently from the learned discriminator in the CVAE experiment, so the comparison between the two should be read as qualitative.

Conclusion

We study how diversity decays under recursive synthetic retraining. We prove that variance in a recursively retrained Gaussian process decays almost surely to zero despite remaining unbiased in expectation, and show through simulation that a verifier causes the expected variance to decay as well, while real-data mixing establishes a stable but reduced equilibrium. We find that this decay behavior persists qualitatively in a higher-dimensional CVAE trained on MNIST, where diversity declines by around 62% across 50 generations with the verifier, and by roughly 41% before plateauing when real-data mixing is added. Without the verifier, fidelity degrades sharply while the Vendi Score rises by 38%. Furthermore, we show that this decline is not reliably captured by standard evaluation metrics, many of which improve over the same period in which diversity and reconstruction fidelity both substantially deteriorate, and Coverage and the Vendi Score move in opposite directions in all three settings. These results indicate that diversity decay follows a predictable, geometric functional form in the Gaussian setting, and that detecting it in practice requires metrics designed to measure diversity directly, tracked alongside a fidelity measure.

Acknowledgements

The author would like to thank Professor Haifeng Xu and TA Bin Yu for their valuable insights and guidance throughout the research process.

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